

Adventures in Programming, Automating, Teaching and Marketing

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- **EvilTester.com**
- **@EvilTester**
- **compendiumdev.co.uk**
- **digitalonlinetactics.com**

You Have Evolved

"I Can Code!"

We have evolved to a point where we can survive in a world of Software

Note: does not need to be programming - any software development role has enhanced survival abilities.

Survive

"We can survive"

- Normal people.**
- Non-programmers.**

May not be able to survive.

Requisite Variety

Cybernetics:

- **"Variety can destroy variety" - Ross Ashby**
- **"Variety can absorb variety" - Stafford Beer**

Thrive

"We don't want to just Survive, we want to Thrive"

```
if( PROGRAMMER & TESTER &  
    ARCHITECT & DESIGNER &  
    AUTOMATOR & MARKETEEER &  
    TEACHER & OPS){  
  
    return SOFTWARE_DEVELOPER;  
}
```

— *You May Not Want to Admit It*

Aspire to be a Software Developer

Getting started is the hardest part.

- Install
- What went wrong?
- What does that error mean?
- That command didn't work what do I do now?
- What to build?

Get People Started

Teach Them the Details Later

Read More code

We used to be exposed to a lot more code

- Mainstream Magazines**
- Multiple Languages**
- Multiple Platforms**
- Aimed at kids**

Bug Detection was for Winners

DEBUGIT

"Simple pleasures . . . are the last refuge of the complex"
— Wilde

You may be 'complex' but whoever wrote this little Spectrumised gem is clearly a mortal of a lesser plain. Actually it was put together, not by Guy Kewney, but by the office glitch, Dick Head, in a rare moment of partial lucidity — usually he finds it difficult just opening the post. Still, just a 'touch' of debugging here and there should do the trick, while at the same time providing some rich moments of simple pleasure to all you nouveau megageniuses who we are proud to call readers.

Help our Dick Head with his hopeless load of programming cobblers (actually, we reckon it contains ten appalling errors) and we in turn promise to smack him across the ear with the full weight of correct entries.

In addition, those extremely kind people at Spectrum (the dealer chain, that is), have agreed to cough up ten prizes each worth £25 — all ready to be spent at one of their vast number of affiliated computer shops. The first TEN correct entries out of the wastepaper basket will be declared the winners (as judged by anyone but DH) and for this great occasion, we could think of no more appropriate date than Friday, January 13 1984.

```
10 BORDER 0: PAPER 0
20 PRINT AT 1,1:"Your Spectrum program puzzle"
30 PRINT
40 PRINT
50 PRINT
60 BORDER 2: PAUSE 3
70 INPUT s#
80 PRINT "Think of a whole number between 0 and 2"
90 INPUT r
100 IF r<>10 THEN GO SUB 350
110 PRINT "Subtract your height (in inches)"
120 INPUT h
130 PRINT "Add your date of birth (ddmmyy)"
140 INPUT dob
150 PRINT "Add your bust/chest size (in centimetres)"
160 INPUT bc
170 IF bc<=90 THEN GO SUB 520
180 PRINT "Subtract your bank account number"
190 INPUT bac
200 PRINT "Add your inside leg measurement (in millimetres)"
210 INPUT il
220 PRINT "Subtract the number you first thought of"
230 INPUT r
240 IF r<>10 THEN GO SUB 350
250 CLS
260 PRINT AT 9,10: FLASH 1:"      WAIT A MINUTE, I'M NOW      EVALUATING #4.1. RE
LEVANT DATA "
270 FOR W=1 TO 50
280 BORDER 4: PAUSE 3
290 PRINT "What's your sex? (m or f)"
300 BORDER 6: PAUSE 3
310 NEXT W
320 CLS
330 GO SUB 430
340 STOP
350 PRINT "NAUGHTY - 'f' has been assumed"
360 PAUSE 75
370 RETURN
380 IF s#="a" THEN GO TO 400
390 RETURN
400 PRINT "Really? What's your phone number?": INPUT ph
410 PRINT FLASH 1:"Thank you. What are you doing tonight?": PAUSE 200: CLS
420 RETURN
430 LET n=INT (RND*10)
440 IF n>3 THEN LET n=n-1
450 IF n=1 THEN GO TO 480
460 IF n=2 THEN GO TO 500
470 IF n=3 THEN GO TO 520
480 PRINT "You are short, fat and ugly, and weigh 32 stones"
490 GO TO 260
500 PRINT "You are stupid and spotty, with severe B.O."
510 RETURN
520 PRINT "You are sylph-like and wonderful, and will soon be rich"
530 RETURN
```

Please send your entries (on paper please) to:
Debugit Competition, Your Spectrum,
14 Rathbone Place, London W1P 1DE.

Polyglot by Default

Osmosis

It wasn't all good though

to see what went wrong. Finally, get rid of the monitor and type in:

SYS 52000

to switch on the hi-res commands properly.

You can then use most of the Commodore 64 graphics programs in *INPUT* without using Simons' BASIC. Whenever you want to reload the hi-res routines, remember to load both sections.

To SAVE anything that you have drawn on the screen do the following POKEs:

POKE43,0:POKE44,32:POKE45,0:POKE46,64

Then SAVE to tape with:

SAVE "filename",1,1

Because *INPUT*'s hi-res graphics program occupies different areas of memory from Simons' BASIC, you may experience some difficulty with routines that save more than one screen within the program itself.

This applies to the paged graphics programs on page 1134 and 1135, and the room designer program on pages 1269 to 1275 and 1308 to 1313.

```
49152 78 a9 0d 8d 08 03 a9 c0
49160 8d 09 03 58 60 20 73 00
49168 c9 40 f0 03 4c e7 a7 20
49176 73 00 a6 7a 86 fb a6 7b
49184 86 fc a0 00 a2 00 dd 00
49192 cf d0 18 e8 a9 0d dd 00
49200 cf f0 29 20 73 00 dd 00
49208 cf f0 f0 a5 fb 85 7a a5
49216 fc 85 7b a9 0d dd 00 cf
49224 f0 03 e8 d0 f6 e8 c8 c8
49232 20 79 00 c0 2c d0 cf a2
49240 0b 6c 00 03 b9 73 cf 85
49248 fb b9 74 cf 85 fc 6c fb
49256 00 00 00 00 00 00 00 00
49264 20 9b b7 e0 10 b0 0c 60
49272 20 fd ae 20 9e b7 e0 10
49280 b0 01 60 a2 0e 6c 00 03
49288 20 70 c0 8e 20 d0 20 78
49296 c0 8e 21 d0 4c ae a7 ff
49304 20 70 c0 8a 0a 0a 0a 0a
49312 85 02 20 78 c0 8a 05 02
49320 85 02 a9 20 85 fe a9 00
49328 85 fd a0 00 91 fd c8 c0
49336 00 d0 f9 e6 fe a6 fe e0
49344 40 d0 ef a9 3b 8d 11 d0
49352 ad 18 d0 29 f0 09 08 8d
49360 18 d0 a2 00 bd 00 04 9d
49368 28 a0 bd 00 05 9d 28 a1
49376 bd 00 06 9d 28 a2 bd 00
49384 07 9d 28 a3 e8 d0 e5 a5
49392 20 a0 00 99 00 04 99 fa
49400 04 99 f4 05 99 e8 06 c8
49408 d0 f1 a9 c8 8d 16 d0 4c
49416 34 c1 ff ff ff ff ff
```

```
49424 20 70 c0 8e f1 cf 20 78
49432 c0 8e f0 cf 20 78 c0 8e
49440 f2 cf a9 d8 8d 16 d0 ad
49448 f0 cf 0a 0a 0a 0d f1
49456 cf 8d f3 cf a9 ff 8d f6
49464 cf 8d f7 cf 8d f8 cf 8d
49472 f9 cf a5 02 8d 21 d0 4c
49480 ae a7 00 00 00 00 00 00
49488 20 73 00 4c 34 c1 00 00
49496 00 00 00 00 00 00 00 00
49504 20 70 c0 8e f6 cf 20 78
49512 c0 8e f7 cf 20 78 c0 8e
49520 f8 cf ad f7 cf 0a 0a 0a
49528 0a 0d f6 cf 8d f9 cf 4c
49536 ae a7 ff ff ff ff ff
49544 20 73 00 a9 15 8d 18 d0
49552 a9 9b 8d 11 d0 a9 c8 8d
49560 16 d0 a5 01 29 fe 85 01
49568 a2 00 bd 00 04 9d 28 a8
49576 bd 00 05 9d 28 a9 bd 00
49584 06 9d 28 aa bd 00 07 9d
49592 28 ab e8 d0 eb a2 00 bd
49600 28 a0 9d 00 04 bd 28 a1
49608 9d 00 05 bd 28 a2 9d 00
49616 06 bd 28 a3 9d 00 07 e8
49624 d0 e5 a2 00 bd 00 d8 9d
49632 28 a4 bd 00 d9 9d 28 a5
49640 bd 00 da 9d 28 a6 bd 00
49648 db 9d 28 a7 e8 d0 e5 4c
49656 7a c2 00 00 82 74 00 ff
49664 20 9b b7 8a c9 00 f0 0d
49672 c9 01 f0 10 c9 02 f0 1b
49680 a6 0b 6c 00 03 a9 15 8d
49688 18 d0 d0 02 a9 17 8d 18
49696 d0 a5 20 2c 11 d0 f0 58
49704 4c 90 c1 a9 3b 8d 11 d0
49712 ad 18 d0 29 f0 09 08 8d
49720 18 d0 a5 01 29 fe 85 01
49728 a2 00 bd 28 a4 9d 00 d8
49736 bd 28 a5 9d 00 d9 bd 28
49744 a6 9d 00 da bd 28 a7 9d
49752 00 db e8 d0 e5 a2 00 bd
49760 28 a8 9d 00 04 bd 28 a9
49768 9d 00 05 bd 28 aa 9d 00
49776 06 bd 28 ab 9d 00 07 e8
49784 d0 e5 a5 01 09 01 85 01
49792 4c ae a7 ff ff ff ff ff
49800 a9 10 2c 16 d0 f0 06 0e
49808 e0 cf 2e e1 cf ad e0 cf
49816 29 07 49 07 8d ef cf ad
49824 e2 cf 29 07 0a 0a 85 fd
49832 ad e1 cf 4a ad e0 cf 6a
49840 4a 4a 85 fb ad e2 cf 4a
49848 4a 4a 85 fc 4a 66 fd 4a
49856 66 fd 18 65 fc 69 20 85
49864 fe a5 fb 0a 0a 0a 90 03
49872 e6 fe 18 65 fd 85 fd 90
49880 02 e6 fe 60 00 00 00 00
49888 20 73 00 4c eb c2 00 02
49896 20 fd ae 20 8a ad 20 f7
49904 b7 a5 15 30 30 c9 01 90
```

```
49912 08 d0 2a a5 14 c9 40 b0
49920 24 20 fd ae 20 9e b7 e0
49928 c9 b0 1a 60 ff ff ff ff
49936 20 fd ae 20 9e b7 e0 03
49944 90 12 a9 10 2c 16 d0 f0
49952 04 e0 05 90 07 68 68 a6
49960 0e 6c 00 03 8e e3 cf 60
49968 ff ff ff ff ff ff ff ff
49976 a9 00 8d ff cf 20 e0 c2
49984 a5 14 8d e0 cf a5 15 8d
49992 e1 cf 8e e2 cf 20 10 c3
50000 4c 58 c3 00 00 00 00 00
50008 20 88 c2 ad e2 cf c9 c9
50016 90 03 4c 53 c4 ad e1 cf
50024 30 0d c9 01 90 0c d0 07
50032 ad e0 cf c9 40 90 03 4c
50040 53 c4 a9 10 2c 16 d0 d0
50048 38 ad e3 cf c9 00 d0 0f
50056 a0 00 ae ef cf b1 fd 3d
50064 a8 cf 91 fd 4c 06 c4 c9
50072 02 d0 0f a0 00 ae ef cf
50080 b1 fd 5d a0 cf 91 fd 4c
50088 06 c4 a0 00 ae ef cf b1
50096 fd 1d a0 cf 91 fd 4c 06
50104 c4 ad ef cf 4a 8d ef cf
50112 ad e3 cf c9 00 d0 06 20
50120 5c c4 4c 06 c4 c9 04 d0
50128 0f a0 00 ae ef cf b1 fd
50136 5d b4 cf 91 fd 4c 06 c4
50144 c9 01 d0 0b 20 5c c4 1d
50152 b8 cf 91 fd 4c 06 c4 c9
50160 02 d0 0b 20 5c c4 1d bc
50168 cf 91 fd 4c 06 c4 20 5c
50176 c4 1d c0 cf 91 fd a5 fe
50184 49 20 85 fe 46 fe 66 fd
50192 46 fe 66 fd 46 fe 66 fd
50200 a9 04 45 fe 85 fe a0 00
50208 ad f9 cf c9 ff d0 1a a9
50216 10 2c 16 d0 f0 25 ad f3
50224 cf 91 fd 18 a5 fe 69 d4
50232 85 fe ad f2 cf 91 fd 10
50240 12 ea ad f9 cf 91 fd 18
50248 a5 fe 69 d4 85 fe ad f8
50256 cf 91 fd ad ff cf f0 01
50264 60 4c ae a7 a0 00 ae ef
50272 cf b1 fd 3d b0 cf 91 fd
50280 60 00 00 00 00 00 00 00
50288 a9 01 8d ff cf 20 e0 c2
50296 a5 14 8d e0 cf a5 15 8d
50304 e1 cf 8e e2 cf 20 e8 c2
50312 a5 14 8d d0 cf a5 15 8d
50320 d1 cf 8e d2 cf 20 10 c3
50328 ad d0 cf 38 ed e0 ef 8d
50336 3c 03 ad d1 cf ed e1 cf
50344 8d 3d 03 b0 0b a9 ff 8d
50352 3f 03 8d 4b 03 4c c2 c4
50360 a9 01 8d 3f 03 a9 00 8d
50368 4b 03 ad d2 cf 38 ed e2
50376 cf 8d 3e 03 b0 08 a9 ff
50384 8d 40 03 4c db c4 a9 01
50392 8d 40 03 ad 3f 03 8d 41
```

```
50400 03 ad 4b 03 8d 4c 03 a9
50408 00 8d 42 03 ad 3f 03 c9
50416 ff f0 0f ad 3c 03 8d 43
50424 03 ad 3d 03 8d 44 03 4c
50432 15 c5 ad e0 cf 38 ed d0
50440 cf 8d 43 03 ad e1 cf ed
50448 d1 cf 8d 44 03 ad 40 03
50456 c9 ff f0 0e ad 3e 03 8d
50464 45 03 a9 00 8d 46 03 4c
50472 39 c5 ad e2 cf 38 ed d2
50480 cf 8d 45 03 a9 00 8d 46
50488 03 ad 43 03 38 ed 45 03
50496 ad 44 03 ed 46 03 b0 26
50504 a9 00 8d 41 03 8d 4c 03
50512 ad 40 03 8d 42 03 ad 43
50520 03 ae 45 03 8d 45 03 8e
50528 43 03 ad 44 03 ae 46 03
50536 8e 44 03 8d 46 03 ad 44
50544 03 6a ad 43 03 6a 8d 47
50552 03 a9 00 8d 49 03 8d 4a
50560 03 8d 48 03 ad e0 cf 8d
50568 d0 cf ad e1 cf 8d d1 cf
50576 ad e2 cf 8d d2 cf ad d0
50584 cf 8d e0 cf ad d1 cf 8d
50592 e1 cf ad d2 cf 8d e2 cf
50600 20 58 c3 ad 47 03 18 6d
50608 45 03 8d 47 03 ad 48 03
50616 6d 46 03 8d 48 03 ad 47
50624 03 38 ed 43 03 ad 48 03
50632 ed 44 03 90 33 ad 47 03
50640 38 ed 43 03 8d 47 03 ad
50648 48 03 ed 44 03 8d 48 03
50656 ad d0 cf 18 6d 3f 03 8d
50664 d0 cf ad d1 cf 6d 4b 03
50672 8d d1 cf ad d2 cf 18 6d
50680 40 03 8d d2 cf 4c 1d c6
50688 ad d0 cf 18 6d 41 03 8d
50696 d0 cf ad d1 cf 6d 4c 03
50704 8d d1 cf ad d2 cf 18 6d
50712 42 03 8d d2 cf ee 49 03
50720 d0 03 ee 4a 03 ad 4a 03
50728 cd 44 03 b0 03 ad 96 c5
50736 ad 49 03 cd 43 03 b0 03
50744 4c 96 c5 ad ff cf c9 01
50752 d0 03 4c ae a7 60 00 00
50760 00 00 00 00 00 00 00 00
50768 a9 02 8d ff cf 20 e0 c2
50776 a5 14 8d bd cf a5 15 8d
50784 bc c6 8e bd c6 20 e8 c2
50792 a5 14 8d be c6 a5 15 8d
50800 bf c6 8e c0 c6 20 10 c3
50808 ad c0 c6 38 ed bd c6 f0
50816 37 90 35 ad bd c6 8d c1
50824 c6 ad bb c6 8d e0 cf ad
50832 bc c6 8d e1 cf ad be c6
50840 8d d0 cf ad bf c6 8d d1
50848 cf ad c1 c6 8d d2 cf 8d
50856 e2 cf cd c0 c6 f0 09 ee
50864 c1 c6 20 98 c4 4c 89 c6
50872 4c ae a7 00 00 00 00 00
50880 00 00 00 00 00 00 00 00
```

```
50888 00 00 00 00 00 00 00 00
50896 a9 01 8d ff cf 20 e0 c2
50904 a5 14 8d a9 c7 a5 15 8d
50912 aa c7 8e ad c7 20 fd ae
50920 20 9e b7 8e af c7 20 10
50928 c3 20 78 c0 8a f0 06 8d
50936 ae c7 20 00 c7 4c ae a7
50944 ad af c7 0a 0a 0a 8d 27
50952 c7 ad af c7 4a 4a 4a 4a
50960 4a 09 d0 8d 28 c7 a2 00
50968 ad 0e dc 29 fe 8d 0e dc
50976 a5 01 29 fb 85 01 bd ff
50984 ff 9d a7 02 e8 e0 08 d0
50992 f5 a5 01 09 04 85 01 ee
51000 0e dc a2 00 8e a7 c7 bd
51008 a7 02 20 4e c7 ae a7 c7
51016 e8 e0 08 d0 ef 60 8d a6
51024 c7 ae ae c7 8e a5 c7 ad
51032 a9 c7 8d ab c7 8d e0 cf
51040 ad aa c7 8d ac c7 8d e1
51048 cf a0 08 8c a8 c7 ad
51056 c7 8d e2 cf ad a6 c7 39
51064 9f cf f0 03 20 58 c3 ad
51072 ab c7 18 69 01 8d ab c7
51080 8d e0 cf ad ac c7 69 00
51088 8d ac c7 8d e1 cf ac a8
51096 c7 8d d0 cf ee ad c7 ce
51104 a5 c7 d0 b3 60 00 00 00
51112 00 00 00 00 00 00 00 00
51120 a9 01 8d ff cf 20 e0 c2
51128 a5 14 8d a9 c7 a5 15 8d
51136 aa c7 8e 97 c8 a9 ff 85
51144 d0 85 0e a9 6a a2 00 85
51152 49 86 4a 8e 9a c8 20 fd
51160 ae 20 b1 a9 a5 6a 8d 9b
51168 c8 a5 6b 8d 12 c8 a5 6c
51176 8d 13 c8 20 10 c3 20 78
51184 c0 8a f0 0c 8d ae c7 20
51192 78 c0 8e 98 c8 4c 06 c8
51200 20 9e b7 4c ae a7 ae 9b
51208 c8 d0 03 4c 94 c8 ae 9a
51216 c8 bd ff ff 8d 9c c8 c9
51224 12 f0 13 c9 92 f0 17 c9
51232 20 b0 2b c9 01 f0 17 c9
51240 02 f0 1b 4c 80 c8 a9 80
51248 8d 99 c8 4c 80 c8 a9 00
51256 8d 99 c8 4c 80 c8 a9 d0
51264 8d 12 c7 4c 80 c8 a9 d8
51272 8d 12 c7 4c 80 c8 29 80
51280 4a 8d 9d c8 ad 9c c8 29
51288 3f 0d 9d c8 0d 99 c8 8d
51296 af c7 8e 9a c8 ad 97 c8
51304 8d ad c7 20 00 c7 ad a9
51312 c7 18 6d 98 c8 8d a9 c7
51320 ad aa c7 69 00 8d aa c7
51328 ae 9a c8 ec 9b c8 f0
51336 06 8e 9a c8 4c 0e c8 a9
51344 d0 8d 12 c7 4c ae a7 00
51352 00 00 00 00 00 00 ff ff
51360 20 73 00 a9 01 8d ff cf
51368 a9 ff 85 0d 85 0e a9 6a
```

```
51376 a2 00 85 49 86 4a 8e e8
51384 cf 20 b1 a9 a5 6a 8d e9
51392 cf a5 6b 8d fa c8 a5 6c
51400 8d fb c8 20 e8 c2 a5 14
51408 8d e4 cf a5 15 8d e5 cf
51416 8e e6 cf 20 10 c3 a0 80
51424 8c eb cf 8c ec cf ac dc
51432 cf 8c ea cf a9 2c 8d 02
51440 ca ae e8 cf ec e9 cf 10
51448 30 bd ff ff 8d e7 cf c9
51456 39 f0 26 c9 30 f0 25 c9
51464 31 f0 4d c9 32 f0 0e c9
51472 33 f0 0e c9 35 f0 15 c9
51480 36 f0 3d c9 37 f0 65 c9
51488 38 d0 03 4c b0 c9 4c e1
51496 ce 4c ae a7 ad ec cf 18
51504 6d d5 cf 8d ec cf ad e4
51512 cf 6d ca cf 20 13 ca 6d
51520 cb cf 8d e5 cf ad eb cf
51528 18 6d d9 cf 8d eb cf ad
51536 e6 cf 6d cf cf 4c d9 c9
51544 ad ec cf 18 6d d6 cf 8d
51552 ec cf ad e4 cf 6d c4 cf
51560 20 13 ca 6d c5 cf 8d e5
51568 cf ad eb cf 18 6d da cf
51576 8d eb cf ad e6 cf 6d cc
51584 cf 4c d9 c9 ad ec cf 18
51592 6d d7 cf 8d ec cf ad e4
51600 cf 6d c6 cf 20 13 ca 6d
51608 c7 cf 8d e5 cf ad eb cf
51616 18 6d db cf 8d eb cf ad
51624 e6 cf 6d cd cf 4c d9 c9
51632 ad ec cf 18 6d d4 cf 8d
51640 ec cf ad e4 cf 6d c8 cf
51648 20 13 ca 6d c9 cf 8d e5
51656 cf ad eb cf 18 6d d8 cf
51664 8d eb cf ad e6 cf 6d ce
51672 cf cd e6 cf f0 08 8d e6
51680 cf a9 20 8d 02 ca ad e7
51688 cf c9 34 30 18 ad e4 cf
51696 8d e0 cf ad e5 cf 8d e1
51704 cf ad e6 cf 8d e2 cf 8e
51712 e8 cf 20 58 c3 ce ea cf
51720 f0 03 4c ec c8 ee e8 cf
51728 4c e6 c8 08 cd e4 cf f0
51736 0a 48 a9 20 8d 02 ca 68
51744 8d e4 cf ad e5 cf 28 60
51752 ff ff ff ff ff ff ff ff
51760 a9 02 8d ff cf 20 e0 c2
51768 a5 14 8d e4 cf a5 15 8d
51776 e5 cf 8e e6 cf 20 e8 c2
51784 a5 14 8d e7 cf a5 15 8d
51792 e8 cf 8e e9 cf 20 10 c3
51800 ad e4 cf 18 6d e7 cf 8d
51808 ea cf ad e5 cf 6d e8 cf
51816 8d eb cf c9 01 30 09 d0
51824 17 ad ea cf c9 40 10 10
51832 ad e6 cf 18 6d e9 cf 8d
51840 ec cf b0 04 c9 c9 90 05
51848 a2 0b 6c 00 03 ad e4 cf
51856 8d e0 cf ad e5 cf 8d e1
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51864 cf ad e6 cf 8d e2 cf 8d
51872 d2 cf ad ea cf 8d d0 cf
51880 ad eb cf 8d d1 cf 20 98
51888 c4 ad ea cf 8d e0 cf 8d
51896 d0 cf ad eb cf 8d e1 cf
51904 8d d1 cf ad ec cf 8d d2
51912 cf ad e6 cf 8d e2 cf 20
51920 98 c4 ad ec cf 8d e2 cf
51928 8d d2 cf ad e4 cf 8d d0
51936 cf ad e5 cf 8d d1 cf ad
51944 ea cf 8d e0 cf ad eb cf
51952 8d e1 cf 20 98 c4 ad e6
51960 cf 8d d2 cf ad ec cf 8d
51968 e2 cf ad e4 cf 8d e0 cf
51976 8d d0 cf ad e5 cf 8d e1
51984 cf 8d d1 cf 20 98 c4 4c
51992 ae a7 ff ff ff ff ff ff
52000 a9 ff 85 33 85 37 a9 1f
52008 85 34 85 38 20 18 e5 a9
52016 06 8d 20 d0 a9 0f 8d 21
52024 d0 a2 00 bd 5d cb d0 d2
52032 ff e8 e0 2e d0 f5 a5 37
52040 38 e5 2d aa a5 38 e5 2e
52048 20 cd bd a9 60 a0 e4 20
52056 1e ab 4c 00 c0 1f 12 20
52064 20 20 20 20 20 20 20 20
52072 49 4e 50 55 54 20 48 49
52080 52 45 53 20 d7 52 41 50
52088 48 49 43 53 20 20 20 20
52096 20 20 20 20 20 20 20 20
52104 20 20 20 ff ff ff ff ff
52112 20 9b b7 e0 08 10 08 8e
52120 ff cf 8a 0a aa 10 05 a9
52128 0b 6c 00 03 bd 31 cc 8d
52136 c4 cf bd 32 cc 8d c5 cf
52144 bd 3d cc 8d c8 cf bd 3e
52152 cc 8d c9 cf bd 35 cc 8d
52160 ca cf bd 36 cc 8d cb cf
52168 bd 39 cc 8d c6 cf bd 3a
52176 cc 8d c7 cf ae ff cf bd
52184 4d cc 8d cc cf bd 51 cc
52192 8d cd cf bd 53 cc 8d ce
52200 cf bd 4f cc 8d cf cf bd
52208 5b cc 8d d4 cf bd 5f cc
52216 8d d5 cf bd 5d cc 8d d6
52224 cf bd 61 cc 8d d7 cf bd
52232 61 cc 8d d8 cf bd 5d cc
52240 8d d9 cf bd 5b cc 8d da
52248 cf bd 5f cf 8d db cf 20
52256 fd ae 20 9e b7 8e dc cf
52264 8a d0 03 4c 9f cb 4c ae
52272 a7 00 00 00 00 01 00 00
52280 00 00 00 ff ff ff ff ff
52288 ff 00 00 00 00 01 00 00
52296 00 00 00 ff ff ff ff ff
52304 00 01 00 00 ff ff ff ff
52312 00 01 00 00 4b 00 b6 00
52320 b6 00 4b 00 4b 00 b6 00
52328 b6 00 00 00 00 00 00 00
52336 20 e0 c2 a5 14 8d e0 cf
52344 a5 15 8d e1 cf 8e e2 cf
```

Learn from Super Stars

JUNE 1984
85p
Dm 5-60

WIN A
TOMY
ROBOT!

**COMPUTER
& VIDEO
GAMES**

EXCLUSIVE!
Program in the
missing screen from
JET SET WILLY!

Grand Prix ace puts racing games to the test!
Games for the Spectrum, CBM 64, BBC, Oric and more

**ANDRÉ'S
NIGHT OFF**

Exclusive!

By Matthew Smith—C&VG
Golden Joystick
Award Winner 1984
Runs on a 16K Spectrum

By now, most of you will have played or seen Matthew Smith's latest weird and wonderful Miner Willy epic from Software Projects, called Jet Set Willy. But did you know that Matthew had an idea which didn't quite make it into the final version of that number one hit game for the Spectrum? We bet you didn't!

Well, Computer and Video Games persuaded our Golden Joystick Award winner, Matthew, to give us an exclusive listing of the missing screen. And now we're passing it on to you — just for the price of this magazine. You can't say we're not generous! Matthew has been slaving away for a couple of weeks to get this game to you — we hope you like it!

As you already know, Miner Willy has been spending the cash he made down the mine and has purchased a vast mansion — complete with a housekeeper and a cook, Maria, the housekeeper, gives Willy a hard time about the mess his guests leave behind after a mammoth party in Jet Set Willy.

In our game, André the cook has waltzed off for the evening without leaving so much as a boiled egg for Willy to munch on! Anyway, our hero marches down to the kitchen to make himself a sandwich and comes face to face with the amazing technicolour man-eating pizzas! GASP!

Can you survive the onslaught of these nasty little doughy monsters? Dare you open Willy's kitchen door? Full instructions are included in Matthew's listing — let us know if you like it!

**ANDRÉ'S
NIGHT OFF**

EXCLUSIVE!

By Matthew Smith—C&VG
Golden Joystick
Award Winner 1984
Runs on a 16K Spectrum

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Now it is all in our browser

Canvas Driver

Points: 1850 Speed: 140 Health: 1 Lives

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Info: OUCH - be careful - Health: 1 - Lives left: 1

@EvilTester

www.compendiumdev.co.uk says

1860 is a HIGH SCORE. Please enter your initials

AAA

Cancel

OK

canvas_driver.html

▶  tpc-check.html

```

104     setTimeout(function () {hideMessage()}, 11000);
105 }
106
107 // DEFAULT KEYS
108
109 var LEFT_KEY = 37;
110 var RIGHT_KEY = 39;
111 var UP_KEY = 38;
112 var DOWN_KEY = 40;
113 var SPACE_KEY = 32;
114 var ENTER_KEY = 13;
115 var ONE_KEY = 49;
116 var TWO_KEY = 50;
117 var THREE_KEY = 51;
118
119
120 var keys = [];
121 keys[LEFT_KEY] = left_key_action;
122 keys[RIGHT_KEY] = right_key_action;
123 keys[SPACE_KEY] = function () {startDifficulty(0)};
124 keys[ENTER_KEY] = promptForValues;
125
126
127 function left_key_action(){
128     carPosition-=1;
129 }
130
131 function right_key_action(){
132     carPosition+=1;
133 }
134
135 function startDifficulty(diff){
136     difficultyLevel=diff;
137     setDifficulty();
138     showInfo(difficultySettings.name);
139     startGame=true;
140 }
141
142 function promptForValues(){
143     promptForValues=true;
144 }
145
146
147
148
149
150 function handleKeyPress(event){
151     // console.log(event.keyCode);

```

{ } Line 1, Column 1

So Much Code Available

- Read Code on Github
- look for codepen (etc.) runnable examples
- YouTube live coding videos
- etc.

People wanted the 'one true book'. Immersion is what it really took.

Writing frameworks is too much fun and should be banned

"Dear Alan,

**I have just started learning
Selenium WebDriver and Java and
I am going to write a framework to
make it easy for anyone to use
WebDriver and Java..."**

"...where do I start?"

"Thanks,

**Someone Who Wants To Save The
World"**

**There is nothing wrong
with wanting to save the
world**

**But don't make that the
first thing you try to do**

What they meant was

- I'm finding it hard to learn this stuff
- I keep making mistakes
- Automating is hard
- It seems easier to write the support code
- I Want a Job For Life

What they needed

- Create solutions to problems
- Learn and gain experience
- Refactor to generalised abstractions
- Build libraries rather than frameworks
- Refactor back to inline code

**Frameworks are
Incredibly hard to build
well and require a lot of
experience**

Dangers of knowing how to code

**Every problem looks like a
reason to create a Tech
Startup...**

**... or a framework, or a
system, or an application,
or code**

Every Problem!

Problems with Twitter

Feed not displayed in time order

Cannot filter tweets - too much noise

What choice did I have?

ChatterScan

.com

@EvilTester

ChatterScan

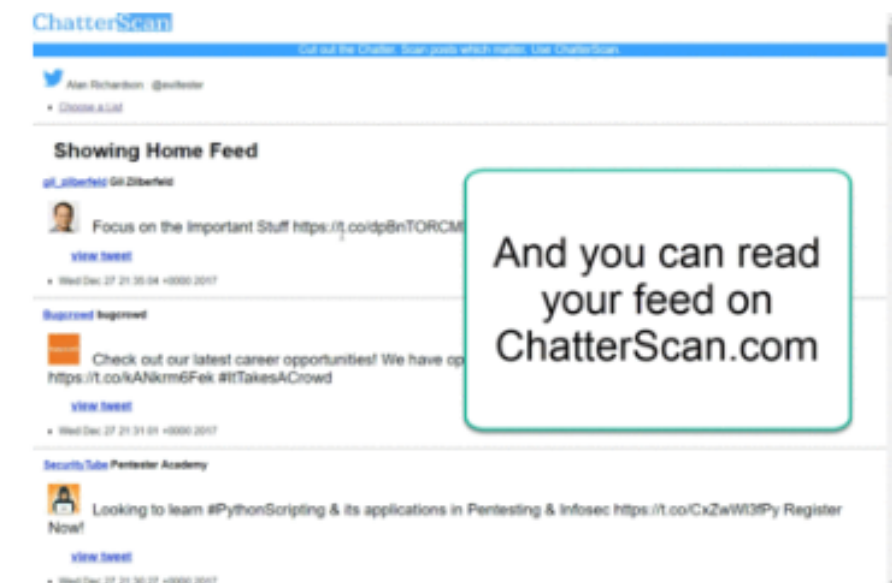
Cut out the Chatter. Scan posts which matter. Use ChatterScan.

Access The Application Here

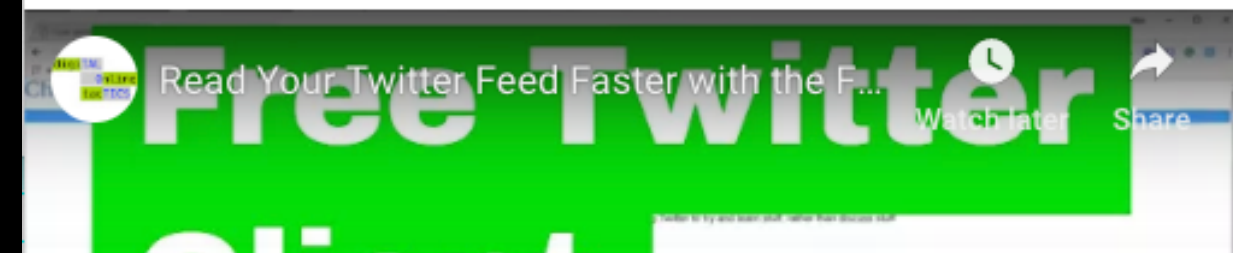
ChatterScan is a minimal Twitter Feed and List reader.

It only shows tweets with links in them. Which is useful when you are using Twitter to try and learn stuff, rather than discuss stuff.

- This app stores none of your data.
- You authenticate using OAUTH.
- We have no access to your passwords, username or email.
- We store none of your tweets.
- The app only has permissions to read tweets, it will not tweet to your timeline, ever.



Demo



Solved Problem with Tools Known at the time

- I did review other tools - Twitter clients, but...
- I found the easiest reliable Twitter API wrapper I could find
- I had an MVP working in hours
- I now have a 'thing'

"Normal" might not get as far as tools.

"Normal" stops at tools.

DANGER - Coder Goes Further

The Ultimate Secrets of Coding (*)...

which I give away while training beginners

— (*) *(some of the secrets)*

**But each secret is a
surface structure**

**Deeper model comes with
experience**

**Applications are just
"code calling other
libraries"**

***The Secret is to find the right
libraries***

But the art is in...

- choosing a library for: easy of use? efficiency? authority?
- abstracting it away or not?
- trusting it, or testing it?

Programmer's Don't Consciously Remember Anything

Google -> StackOverflow

But the Art is in ...

- wrapping the copied code in a test
- refactoring it into our domain
- taking responsibility for it

IDEs write and fix code

Press Alt+Enter repeatedly, don't remember syntax.

Control+Space and Code completion, don't try and remember library functionality.

But the art is in...

- the design, not the syntax
- writing what we want to see

**Programmers Don't
Actually Run the App to
see if it works**

Unit Tests

Continuous Integration

But the art is in...

- creating a trust building assertion safety net**
- creating a robust execution mechanism**

**Everything you need to
know you can learn from
writing Text Adventures**

Coding Them



You Learn

- Parsing, Tokenisation
- Domain Specific Languages
- Writing Interpreters
- Data Generation
- Data Structures
- Data Compression
- Engines
- Incremental releases, System, MVP

No Limit

- Web Apps
- REST API
- Multi-User

You Look.

You are aware that:

- The Wizard Says: Welcome everybody to the test game

Look

You are in: The Central Test Room

This is the room in the center of the test game

You notice:

- A sign on the wall
 - [Examine it](#)

Visible Exits that you can see:

- [S](#)
- [E](#)
- [W](#)
- [N](#)

You can see:

You can also:

- [look](#)
- [inventory](#)
- [help](#)

Coding Not Always Required

- finished games
- using an engine
- data driven
- learn design
- DSL creation



Playing Them

- Architecture Mapping
- Testing
- Exploring
- Reverse Engineering



Marketing

- Solve Problems
- Solution Development

Solution Development

- Code for individual joy and fun**
- Survival not the end goal**
- Thrive as a Software Developer**
- Build Solutions rather than Software**
- Help other people Thrive**

Aspire to be a Solutions Developer

End

Alan Richardson

EvilTester.com

<https://twitter.com/eviltester>

Games shown

- <https://www.compendiumdev.co.uk/games/buggygames/>
- <https://www.compendiumdev.co.uk/page/restmud>

References

- various screenshots of magazines from archive.org
- MUD MUD: Messrs Bartle and Trubshaw's Astonishing Contrivance
 - <https://www.youtube.com/watch?v=YfX8Z8W9JCE>

About Alan Richardson

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